

KAMAL PATEL

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EDUCATION

Rutgers, The State University of New Jersey · New Jersey

September 2022 - June 2026

MS & Doctoral studies in Industrial & Systems Engineering (**Completed two years of PhD courses**)

Courses: Production Analysis, Supply Chain Engineering, Linear Programming, **Lean Six Sigma (Rutgers Business School)**, Regression Analysis, Systems Reliability, Data Mining II, Time-Series Analysis.

Certification: Lean Six Sigma Green Belt (CFQ International LLC)

PROJECTS

Safety Stock and Inventory Simulation Engine | Python, NumPy, Pandas, SciPy, Matplotlib

[GitHub](#)

- Formulated parametric statistical models using SciPy to evaluate safety stock requirements, optimizing trade-offs between 95%+ customer fill rates and inventory holding costs.
- Engineered discrete-event simulation in python to model multi-SKU inventory dynamics, reorder points, and EOQ under demand and lead-time volatility.
- Developed vectorized NumPy decision models to simulate 100+ policy combinations (ROL, ROQ), identifying minimum-cost operating points while preventing stockouts.
- Executed trade-off analysis across 50 SKUs using Pandas aggregation pipelines, proving via sensitivity curves that lead-time variance exerts a higher cost impact than daily demand variance.

E-Commerce Delivery Optimization Analytics | MySQL

[GitHub](#)

- Analyzed 100K+ order fulfillment records in MySQL to evaluate baseline On-Time In-Full (OTIF) performance and pinpoint root causes of network-wide delivery latency.
- Leveraged CTEs and Window Functions to construct performance rankings that identify the highest-risk product category per state.
- Quantified localized revenue exposure by writing CTE-driven SQL logic to aggregate high-value SLA breaches (>\$200 value, High/Critical priority) across states.

End-to-End Inventory & Supply Chain Performance Dashboard | Power BI (DAX)

[GitHub](#)

- Developed an interactive Power BI dashboard tracking core supply chain **KPIs**, including Warehouse Utilization Rate, Days Sales of Inventory (DSI), and Inventory Turnover Ratio.
- Built an interactive Power BI dashboard querying multi-warehouse inventory datasets to track SKU turnover, safety stock, and reorder alerts.
- Modeled relational inventory datasets to deliver actionable recommendations for inventory replenishment and stocking strategy.

TECHNICAL SKILLS & LEADERSHIP

Data Analytics Tools/Packages Python (Pandas, NumPy, Scikit-learn), PyTorch, R, MATLAB, SQL, Minitab

Software Packages Flexsim, GurobiPy, AnyLogic, MS Project, Visio, AutoCAD

Leadership Lead Mechanical Engineer (NYU **International Ground Vehicle Competition**)

ACHIEVEMENTS/CERTIFICATIONS

- **Data Mining II** Rutgers University ([Best Presentation](#))
- **Regression Analysis** Rutgers University ([Best Project Performance](#))
- **SQL for Data Science** University of California, Davis ([Certificate](#))

WORK EXPERIENCE

Rutgers University - Industrial & Systems Engineering

Jan 2025 – June 2026

Graduate Teaching Assistant

Piscataway, NJ

- Assisted 30+ students on linear programming concepts and mentored optimization projects using Excel Solver.
- Facilitated weekly recitations for reliability engineering course and guided students through MATLAB-based computational assignments and analysis.

Kismet Technologies

May 2023 - Aug 2023

Industrial Engineer Intern

Orlando, FL

- Customized Katana MRP software for inventory management, driving a **20%** improvement in inventory accuracy.

- Revamped batch manufacturing data tracking by building customized PowerApps application, boosting operational efficiency by **11%**.
- Initiated production schedules and authored standard operating procedures (SOPs) for batch acceptance testing and to ensure ISO 9001 compliance.

Mareana: Industry Consulting

Nov 2022 - Dec 2022

Project Head

New Brunswick, NJ

- Spearheaded a 4-person consulting team analyzing production line efficiency for a turbine manufacturer.
- Engineered a **FlexSim discrete event model** to optimize line balancing, reducing production cycle time by **23.5%**.
- Analyzed production data and proposed a solution with a projected throughput increase by **10–12%**.

Susha Founders and Engineers

Dec 2021 - May 2022

Manufacturing Engineer Intern

Gujarat, India

- Conducted time-and-motion studies across assembly stages for 2 production processes, reducing cycle time by 8%.
- Analyzed historical inventory datasets using SQL, VLOOKUPS, and Pivot Tables to forecast demand and establish new EOQ and reorder points.
- Enforced 5S principles and optimized monthly Overall Equipment Effectiveness (OEE) tracking, yielding a ~10% operational efficiency gain.